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Mobile demonstration unit for fast- and catalytic pyrolysis: The combustion reduction integrated pyrolysis system (CRIPS)☆

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ABSTRACT

A mobile pyrolysis unit, designed to process biomass at a rate of 83.3 kg h^{-1} (two metric tons per day, MTPD) has been developed, constructed and its operation demonstrated for on-farm or in-forest production of pyrolysis oil (bio-oil). The trailer-mounted pyrolysis apparatus is based on the combustion reduction integrated pyrolysis system (CRIPS), a patented, dual fluidized bed, biomass pyrolysis design developed by USDA and the University of Pretoria. The mobile design goal was to demonstrate efficacy of on-farm production and coordinate station-to-station operation to simulate a collective system of several unit operations within a distributed/satellite system of pyrolysis biorefinery. Key design features that provide utility for remote operations including in-situ (on-trailer) generation of heat and electric power required for self-sufficiency were tested at this scale. Extensive trials on three feedstocks important to US agriculture, namely woody biomass, switchgrass and horse litter, were conducted. The results show that a processing rate of up to 40 kg h^{-1} (approximately 1 MTPD) was successful and easily achievable for the said biomass on the trailer; however, beyond this rate operational problems such as pressure imbalances between the dual-bed reactors were encountered, hampering process control. The operation of the CRIPS under catalytic pyrolysis conditions was also successful with catalyst regeneration readily achievable with the design. Bio-oil yield for neat/traditional and catalytic pyrolysis were in the 45% and 5–10% ranges respectively, similar to lab scale, demonstrating production of large volumes of bio-oil on-farm. Bio-oil quality for catalytic pyrolysis was consistent over several hours on stream due to continuous catalyst regeneration that the design affords yielding high levels of BTEX compounds; however, deactivation due to alkali contamination was noticeable at cumulative biomass to catalyst ratios of > 6/1. Compared with laboratory scale results, non-catalytic pyrolysis product quality for the demonstration scale wavered between that of regular pyrolysis oil produced under nitrogen atmosphere and that of the tail-gas reactive pyrolysis (TGRP) as partial recycle of effluent gas was possible with the mounted CRIPS design. With successful demonstration at 1 MTPD capacity the mobile system meets the technology readiness level for pre-commercial design suggested for blueprints (*i.e.* development of full-scale prototype), however more testing is underway to remove the bottlenecks impeding operation at the full-scale design.

1. Introduction

Biomass pyrolysis, the thermal decomposition in the absence of an oxidant, yields bio-oil (pyrolysis oil), biochar, and non-condensable gases (NCG). Fast pyrolysis is the type of pyrolysis achieved by rapid heating of finely-divided biomass to induce high heat rate at short contact times. At temperatures around 500–550 °C in an oxygen-free atmosphere, fast pyrolysis maximizes the production of bio-oil by separating the bio-char, and rapidly quenching and condensing the effluent gas and bio-oil vapors [1–3]. Bio-oil has significant potential as precursor or as an intermediate feedstock for the production of advanced bio-fuels and renewable chemicals. Dubbed liquid biomass, it is significantly denser than its parent biomass. This attribute, combined with the relative simplicity of the process of making it, has led to the idea that distributed scale processing on-farm for the conversion of biomass to bio-oil could be an economically viable and environmentally sustainable biorefining concept. Densification of biomass at the biomass source (at- or near-the-farm, or in-forest) can save significant costs by transporting the liquid, rather than the raw biomass material, to a centralized facility for further refining. Densification of biomass via the

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Nomenclature

Pertinent Notations

ASTM	American Society for Testing and Measurement
BFB	Bubbling fluidized bed
BRDI	Biomass research and development initiative
BTEX	Benzene, toluene, ethylbenzene and xylenes
CFP	Catalytic fast pyrolysis
CRIPS	Combustion reduction integrated pyrolysis system
ESP	Electrostatic precipitator
FCC	Fluid catalytic cracking
GC	Gas chromatography
GC/MS	Gas chromatography/mass spectroscopy
HHV	Higher heating value
HZSM	Zeolite socony mobil-5
LPG	Liquefied petroleum gas
MSD	Mass selective detector
MTPD	Metric ton per day
NCG	Non-condensable gas
NIFA	National Institute for Food and Agriculture
PDU	Process development unit
PRV	Pressure regulating valve
SCADA	Supervisory control and data acquisition system
TCD	Thermal conductivity detector
TGRP	Tail gas reactive pyrolysis
USDA	United States Department of Agriculture
WHSV	Weight hourly space velocity
US-DOE	United States Department of Energy
PP20	20kW Power Pallet

use of mobile or distributed torrefaction or slow pyrolysis systems to produce bio-char on site has been suggested by technoeconomic assessments [4–6]. The economic viability of such a scenario using fast pyrolysis technologies to produce bio-oil has also been supported by several hypothetical studies [5–10], but as of yet has been unverified with actual on-farm demonstration scale data. To address this gap, a skid-mounted, mobile pyrolysis demonstration unit was constructed to test the feasibility of reaching the biomass source followed by on-the-farm bio-oil production to simulate the distributed/satellite processing model. Few other mobile fast pyrolysis systems have been tested to date [11]. The mobile design for the unit described herein, follows the combustion, reduction integrated pyrolysis system (CRIPS) design [12] which encompasses autothermal processing for energy self-sufficiency and allows for the testing under various on-farm or in-forest situations with a single demonstration unit rather than several units in the satellite system.

As previously mentioned, fast pyrolysis requires rapid heating to maximize bio-oil production. While many different types of reactors and heating schemes can be effective in achieving this condition, the bubbling fluidized bed (BFB) has found a widespread application in the process industry because of its high heat transfer rates and short vapor contact times. In most fluidized-bed pyrolysis designs, entrained bio-char is easily removed from the gas and bio-oil vapor stream with a cyclone separator prior to quenching with liquid/gas heat exchanger(s) where most of the aqueous fractions of the bio-oil is condensed [1,13,14]. Typically, effluent gas from the condenser(s) still contains much of the condensed, non-aqueous bio-oil as aerosols which are often extracted by an electrostatic precipitator (ESP) downstream. A schematic of a process flow diagram depicting BFB fast pyrolysis reactor, cyclone separator, condenser, and ESP is depicted in Fig. 1, following our laboratory (USDA) bench scale unit [14]. This diagram depicts the minimum and complete assembly of auxiliary unit operations one would need in a primary pyrolysis system. Such includes a feeding system comprising a feed hopper with metering auger, a rotary airlock, and an injection auger.

When it comes to end use, utilization of fast pyrolysis bio-oil has been limited because of the poor fuel and handling properties of the bio-oil. Its chemical composition, characterized by high concentrations of compounds containing reactive oxygenated functional groups, makes it thermally unstable and challenging to refine to transportation fuels. For the same reason, it also has a lower energy content (enthalpy of combustion) than fossil fuels and is not seamlessly compatible with most industrial furnace or boiler fuels. Improvements to basic fast pyrolysis processes that would result in higher quality, partially deoxygenated bio-oil have therefore been the research goal needed for the success of pyrolysis biorefineries. Some of the promising technologies have included catalytic fast pyrolysis (CFP) [15–22] and tail gas reactive pyrolysis, a USDA invention [23,24] wherein a portion of the process tail gas (NCG gas) is recycled back to the BFB reactor to replace some of the nitrogen employed as the fluidizing gas. Each of these processes has demonstrated increased C/O ratios (decreasing oxygen content) in the oil phase of the bio-oil collected at the ESP. To accomplish the recycling of the tail gas a recycle blower and a pressure regulating valve (PRV) are used whereby the recycle blower increases

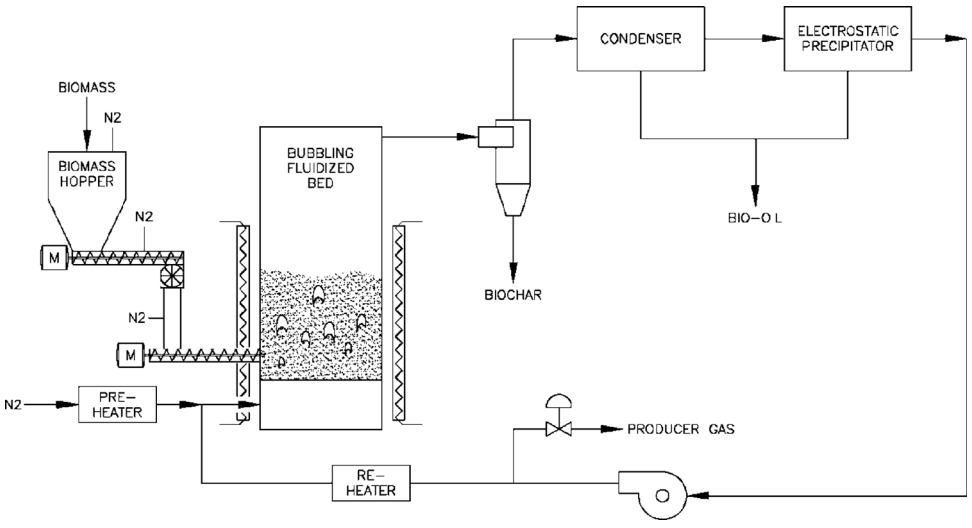


Fig. 1. Typical Fast Pyrolysis System with a Bubbling Fluidized Bed & Gas Recycling loop for Tail Gas Reactive Pyrolysis.

the pressure of the tail gas in order to introduce it to the BFB; the PRV is used to regulate the system pressure in conjunction with the fluidizing nitrogen stream.

While the use of a heating mantle to introduce heat into the BFB reactors works quite well at laboratory or bench-scale units, scaling to near commercial size will require fundamental changes to the heating scheme required for these larger BFB reactors as the reactor surface area to volume ratio decreases with increasing volume (assuming geometric similarity when scaled). As the surface area to volume ratio decreases, higher temperatures are required to ensure sufficient heat transfer into the larger volume. However, increased wall temperatures can drive the pyrolysis reactions away from optimal conditions needed for producing the bio-oil. The CRIPS [12] design addresses some of these challenges including scale-up limitation with the ability to operate in the traditional fast pyrolysis, TGRP and CFP modes and hence lends itself as a logical design basis for the mobile system.

2. Experimental

2.1. Mobile CRIPS reactor design

The goal of the mobile system was to be deployed at remote areas, hence heat integration and energy self-sufficiency was the primary concern. It also has to produce high-quality, partially-deoxygenated bio-oil at the farm site that can be shipped off-site, with any biochar produced left on the fields to amend the soil. The CRIPS was therefore the logical choice for the design of the mobile device as it has all the aforementioned attributes including use of on-board generated energy

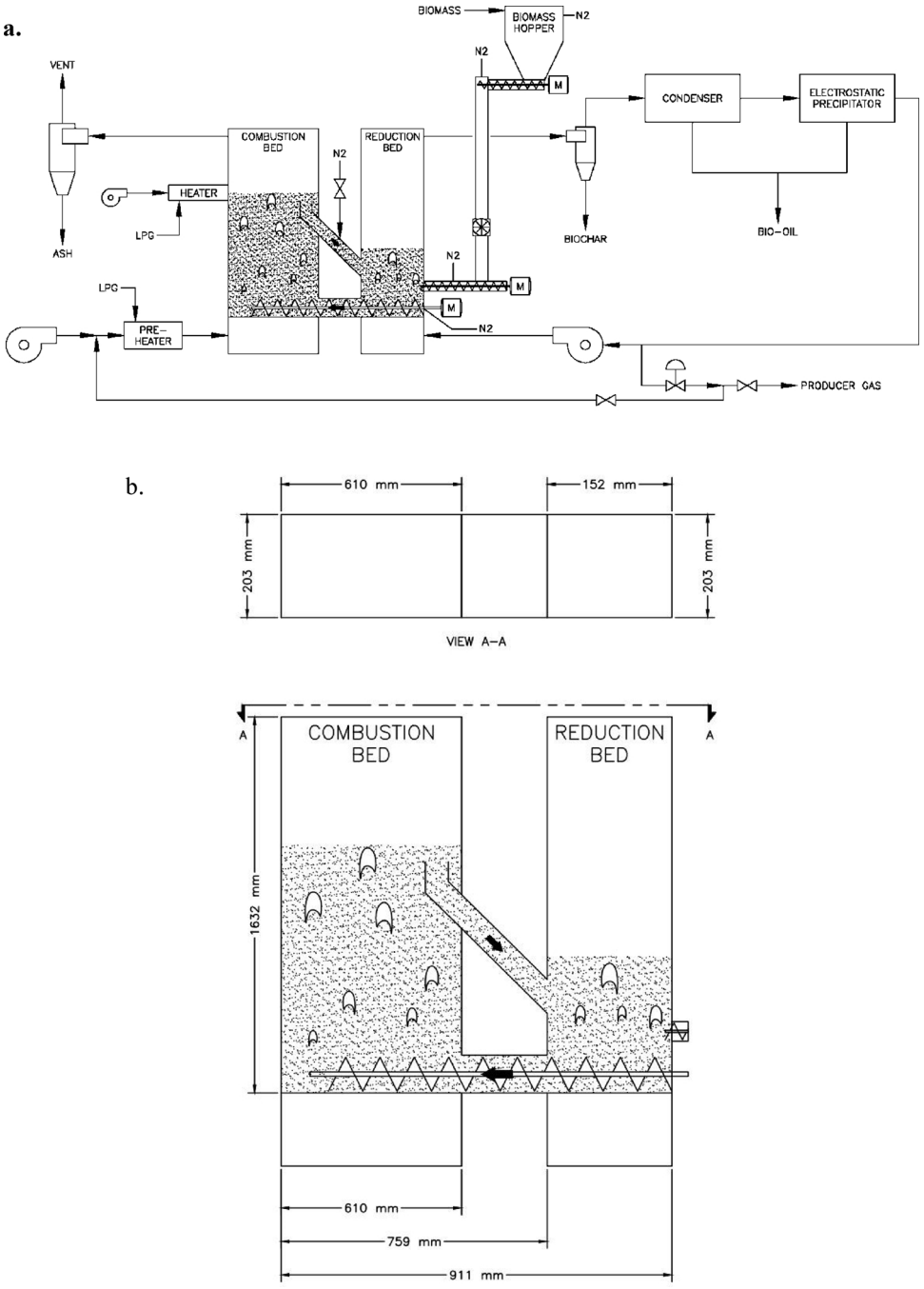


Fig. 2. Schematic of the Combustion Reduction Integrated Pyrolysis System (CRIPS). a. Full view; b. Twin-bed diamensions.

to power the system in remote areas. Fig. 2 presents the CRIPS arrangement in its primary form as patented. It has been described elsewhere [12] but briefly, the key design feature is an incorporation of a twin fluidized bed reactor system into the regular TGRP process flow setup [23,24].

The reactor system consists of two fluidized bed reactors (Fig. 2) in which one functions as a pyrolyzer (a reducing bed created by fluidized exhaust gas medium) and the other as a combustor (an oxidizing bed created by fluidized air medium) with the latter providing the needed heat generated from the combustion of char returned from the pyrolyzer for the endothermic pyrolysis reaction in the former. The heat exchange is accomplished by circulating the solid medium (inert sand or catalyst) acting as the "heat flywheel" similar to the fluid catalytic cracking (FCC) process used in petroleum refineries. As shown in Fig. 3, the twin reactor and downstream auxiliary equipment are skid-mounted on a trailer within a footprint of 11.8 m long × 2.5 m wide × 3.7 m high (38' 8" long × 8' 4" wide × 12' 3" high) having an estimated weight of 5900 kg (13,000 lbs). Electrical energy is generated on the trailer by a Power Pallet 20kw (PP20) gasifier/generator system (Figure S1) (All Power Labs, Berkeley, California), fueled by wood chips or pellets, and mounted on the trailer. Table 1 lists the utilities required for operation on the mobile CRIPS which comprise electric motors, LPG, inert gas for blanketing and the 2PP20 gasifier/generator allowing for a complete off-grid operation. The condensers downstream of the twin-bed reactors are cylinders, 2.1 m high × 0.211 m in diameter wrapped with a 0.323 m OD water jacket. Cooling water is circulated through the condensers in series countercurrent to the pyrolysis gas flow. The gas exiting the final condenser is typically around 20–50 °C, depending on ambient temperature. Cold water for the condensers is supplied by a 15 kW on-board Thermoflex TF15000 recirculating chiller (Thermo Fisher Scienfific, Waltham, MA) that can circulate up to 75 dm^3 of cooling water at up to 240 kPa with the capability of cooling down to 5 °C. Combustion and pyrolysis bed blowers are two-stage regenerative blowers (also known as a side-channel blowers) powered by external electric motors and driven via belt and pulleys. The combustion bed blower is a model RB63 (All-Star Products, Memphis, TN) driven by a 10HP (7.46 kW) motor. The pyrolysis bed blower is a model RB63 (All-Star Products, Memphis, TN) driven by a 5 HP (3.73 kW) motor. Fig. 4 depicts an aft-view section through the twin BFB reactors on the trailer in a fashion similar to that depicted Fig. 3. However, note that the orientations of these two figures are left-to-right reversed.

Table 1
CRIPS Utility Loads.

Utility	Details
Electricity	208 V, 3 phase Typ. 10-16 kW, depending on LPG vaporizer and condenser chiller demand. Can be provided by gasifier/generator fueled by various biomass materials (e.g. wood chips)
Propane	Typ. 36 standard liters per minute (slpm) LPG for heat-up Typ. 2.5–5 slpm operating (2.5 slpm for preheater safety pilot)
Inert Gas	Typ. 55 slpm @ 40 psig (2.8 barg) Nitrogen was used for all trials in for this project

2.2. CRIPS operation

Both pyrolysis and combustion reactors were fluidized beds of inert (silica sand between -20 + 25 US mesh or ~ 650 μm) or heterogeneous catalyst (HZSM-5 zeolite in the 700 μm particle size range) with minimum fluidization velocities calculated at 15.2 and 18.3 cm/s respectively. As shown in Fig. 4, circulation of the fluidized media between the twin BFBs is accomplished by the following path; first, the sand auger (7) withdraws the fluidized media from the reduction bed

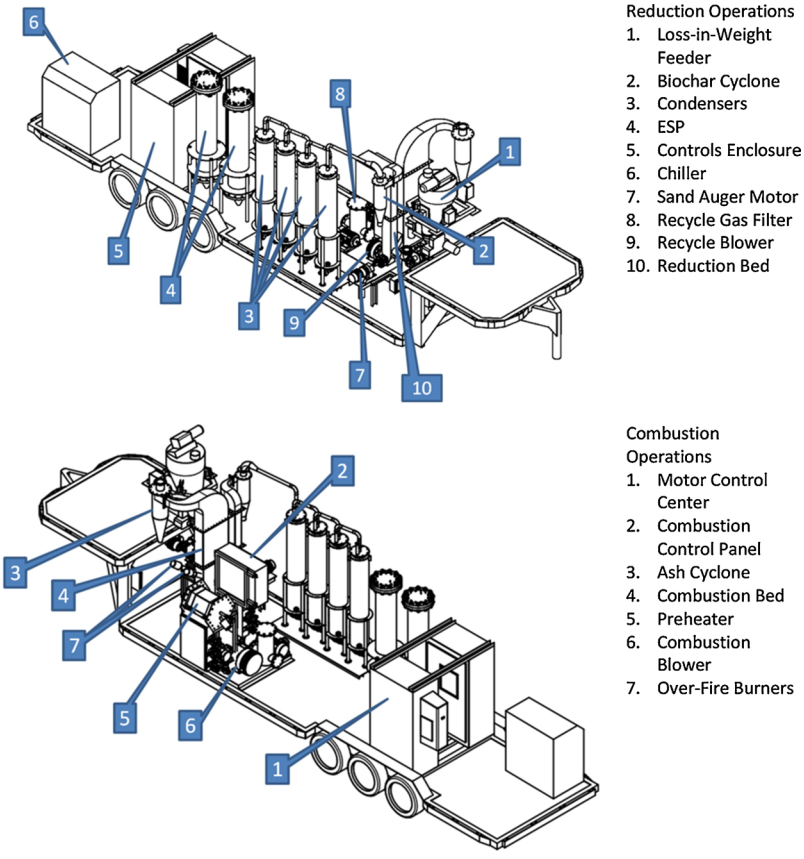


Fig. 3. CRIPS Trailer Unit Layout.

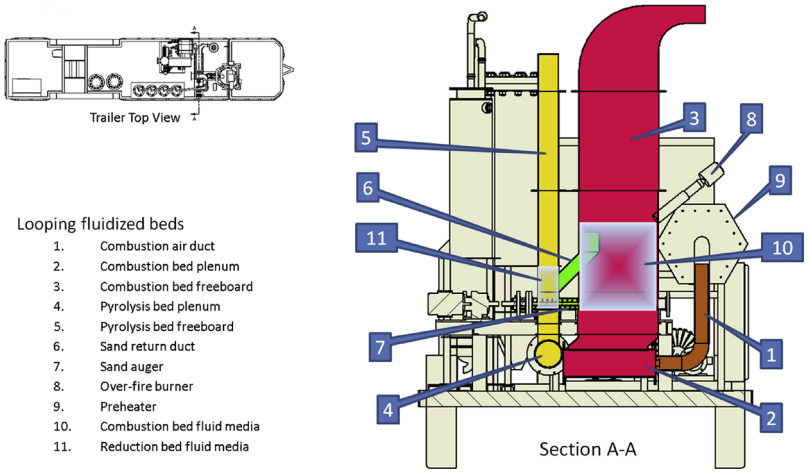


Fig. 4. Aft view depiction of CRIPS reactors and burner assembly.

(11) and pushes it into the combustion bed (10); second, as the media accumulates in the combustion bed any excess volume of the fluidized media overflows the top of the return duct (6) and is returned to the reduction bed. Preheated air from the pre-combustor (9) is introduced to the combustion bed plenum (2) via the combustion air duct (1). This flow of air expands and fluidizes the solid media in the combustion bed where it can be efficiently heated by the over-fire burners (8), combusting any entrained biochar and/or biomass, elutriating fine ash particles, and exiting the system via the combustion bed freeboard (3). Likewise, recycled gas from the ESPs is returned to the pyrolysis bed plenum (4), fluidizing the solid media in the reduction bed (11). A mixture of recycled gases, entrained biochar particles, and bio-oil vapors exit the reduction bed via the pyrolysis reactor freeboard (3).

The combustion bed replaces the heating mantle in the two aforementioned fast pyrolysis processes, and also serves as a coke combustor for constant regeneration of catalysts if used as the solid media for CFP operation. Fuel is combusted upstream and within this BFB to efficiently heat the fluidizing media, typically to ca. 100–150 °C above the desired reduction (pyrolysis) bed average temperature. The heated fluidizing media is transferred into the reduction bed, the BFB where the pyrolysis reactions occurs. The cooled, fluidizing media is then returned to the combustion bed for reheating. The circulation of the fluidizing media is like a FCC system with the combustion bed analogous to an FCC regenerator and the reduction bed analogous to a FCC reactor. Circulation of the solid medium among the two BFBs is controlled by a common auger connecting the two BFB reactors whereby the circulation rate is increased with increased auger rotational speed. Due to the higher temperature maintained in the combustion bed, this increased medium circulation will result in an increased reduction bed temperature.

For start-up when the system is cold, and biomass is not being processed, the combustion bed is fueled by liquefied petroleum gas (LPG). This LPG is burned within the lean phase of the combustion bed and the preheater upstream of the combustion bed, producing the heat required to raise the temperature of the fluidizing solid media suitable for operating the CRIPS. When the CRIPS is processing biomass, the fuel used to maintain the temperature of the combustion bed comes from several sources: LPG, entrained biochar & biomass returning from the reduction bed to the combustion bed, and surplus producer gas generated by the CRIPS. For most biomass materials tested including hardwoods, switchgrass, and equine waste (horse litter) the entrained biochar and biomass contain sufficient calorific energy to maintain the temperature of the combustion bed allowing for the LPG flow to the CRIPS to be discontinued and freeing the producer gas for other purposes.

To operate the system, pyrolysis temperatures were maintained between 500–550 °C, with the combustion bed temperatures maintained between 650–750 °C. Up to about 41 kg h^{-1} of various biomass feedstock was fed into the reactor as established by the loss-in-weight feed system. To maintain system balance and continuous feeding at this feed rate approximately a total of 40 L/min N_2 from gas cylinders stored on-board the trailer was added to serve as fluidization make-up gas, flow control and feed purging gas. A supervisory control and data acquisition system (SCADA) system was developed to act as the operator interface and allow for control of the CRIPS via a programmable logic controller. The SCADA system was developed in Siemens PCS7 and runs on a Siemens Simatics PCS7 Box. Field devices were either directly connected to the 'Fieldbus' (running a combination of Profibus PA and DP) or a single Siemens ET 200 distributed I/O controller.

While the CRIPS is in operation, a vapor slip stream is continuously withdrawn from the reduction bed plenum and analyzed for CO, CO_2, H_2, CH_4, C_2H_4, C_2H_6, and C_3H_8 by an Inficon Fusion Micro GC dual column gas analyzer equipped with thermal conductivity detectors (TCD).

2.3. Catalytic pyrolysis CRIPS operation

HZSM-5 catalysts (CBV8014, SiO_2/Al_2O_3 = 80) were provided as extruded pellets by Zeolyst International (Conshohocken, PA). The pellets contain approximately 20 wt% of alumina binder material. The pellets were crushed to particle size range of 500–1180 μm (avg. particle size ~ 700 μm) prior to use. For catalytic pyrolysis operations, the CRIPS reactor was loaded with 12.3 kg of this material in place of the sand as the fluidizing medium used for the non-catalytic experiments. Feeding of switchgrass was started when the pyrolysis bed reached 500 °C; typically the combustion bed temperatures were in the 600–650 °C range at this point. Biomass was fed at 10 kg/h in the catalytic pyrolysis experiments. Upon feeding of the biomass temperatures between 500–550 °C were maintained in the pyrolysis bed, but temperatures in the combustion bed rose to the 750 °C–800 °C range, resulting from combustion of the coke on the catalyst as it is circulated. Experiments were run for about 3.25 h. Post run the catalyst was removed from the bed and weighed. After 195 min of operation, the particles had been reduced in size and 1.9 kg (15.4%) of catalyst was lost to attrition of fines that exited the reactor with bio-char on the pyrolysis side or via the chimney on the combustion side. Previous to the subsequent run fresh catalyst was added as makeup. The Brønsted acid site densities of the HZSM-5 fresh and spent catalysts were determined by reactive gas chromatography utilizing isopropylamine decomposition following a method modified from Abdelrahman et. al. (modified to use MS detection) [25].

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2.4. Biomass, product and catalyst characterization

As mentioned earlier the CRIPS was tested on three feedstocks important to US agriculture, switchgrass, woody biomass (hardwood) and horse litter. Hardwood biomass was provided by Mesa Reduction Engineering, Inc (Auburn, NY). Horse litter was provided by Morrisville State College (NY), from its equine rehabilitation facility. Switchgrass was provided by the McDonnel Farm (East Greenville, PA). Elemental analysis (C, H, N, O), water and ash content of the feedstocks (method below) is provided in Table 2. Feed material was typically ground to 2–3 mm particle size before use.

Elemental analysis was performed using a Thermo Flash EA1112 CHNS/O analyzer by complete combustion of the material followed by GC quantification of combustion products. Oxygen was determined by difference after accounting for CHNS, ash and water. The water content of bio-oil fractions was measured using Karl Fischer titration in methanol with hydranal Karl Fischer Composite 5 (Fluka) as titrant. Quantification of pyrolysis oil compounds by Gas Chromatography with Mass Spectroscopy detection (GC/MS) was performed using a Shimadzu GC-2010 equipped with a mass selective detector (MSD). The column used was a DB-1701 at 60 m × 0.25 mm and 0.25 μm film thickness. The oven temperature was programmed to hold at 45 °C for 4 min, ramp at 3 °C/min to 280 °C and hold at 280 for 20 min. The injector temperature was 250 °C, and the injector split ratio was set to 30:1. The flow rate of the He carrier gas was 1 mL/min. The pyrolysis oil samples for GC analysis were prepared as 3 ± 1 wt% solutions in acetone. For quantification of individual pyrolysis oil compounds, response factors relative to the internal standard fluoranthene were determined using authentic compounds.

3. Results and discussion

We begin by illustrating system operability and integrity as it responds to pressure and temperature beginning from heat-up through feeding and eventual conversion to pyrolysis products. Fig. 5 presents pressure (Fig. 5a) and temperature (Fig. 5b) trending with time on stream with switchgrass as feedstock for one of many test-runs at our laboratories in Pennsylvania. Other process trends such as fuel rate from start-up until auto-thermal conditions as well as power draw are presented in the supporting information.

As noticed for this example run, the system pressure stabilizes 1.5 h after the heat-up process is initiated which is enabled using LPG but system temperatures continue to rise until 2.5 h of ramping when the set-point temperature for the pyrolysis bed (500 °C) is reached. The bed temperature is allowed to stabilize for 30 min at around 500 °C until feed is introduced. At this point combustion bed temperature is sustained at around 700 °C. Upon start of the chiller to provide cold water to the condenser (8:45) the system's power draw jumps twofold (Fig. S1) and another small increase is noted at 10:00 with start of the feed augers for biomass feeding. Both the reactor plenum static pressures as well as the differential pressure across the combustion bed remain steady but the differential pressure across the pyrolysis bed begins to fall and as a result the pressure differential between the two reactors rises. Complete equilibrium is reached 2 h thereafter where the process can be classified as a full steady-state operation i.e., some 5 h after start-up although bio-oil collection had begun 20–30 min upon feeding. At this time the pressure differential between the combustion and the pyrolysis chambers is steady at about 25 mbar (~10 in wc) and the combustion bed and chimney temperatures rise as biochar, introduced into the combustion bed, is combusted maintaining a temperature difference of 150 °C between the beds. At this time the propane is switched off and the system is in full auto-thermal mode for about 2 h until the run is terminated, and the system is shut down (Figures S3 and S4).

At the full steady-state condition, the pyrolysis chimney is about 400 °C when the pyrolysis temperature is 500 °C, a temperature difference of 100 °C, while the combustion bed and chimney temperatures are about the same. Importantly, there are no temperature differences within various sections of the both pyrolysis and combustion beds. The agreement of the bed temperatures indicates close to isothermal bed, an indication of a well fluidized bed. The steady pyrolysis temperature indicates successful control enabled by effective recirculation of the solid medium between the twin-bed with the use of the return auger. The high combustion bed chimney temperature of more than 800 °C suggests there is sufficient waste energy available that may be recuperated for various auxiliary applications such as feedstock drying.

4. Conventional fast pyrolysis operation

Table 3 presents the yields of recovered products for switchgrass, hardwoods and horse litter from conventional fast pyrolysis using CRIPS. This arrangement follows the traditional fast pyrolysis mode of operation for the production of bio-oil and is carried out using silica sand with nominal diameter of 655 μm as the fluidizing and heat transfer medium. Total liquid yields obtained were about 48.5 wt% of the input feedstock material for either switchgrass or hardwood. It must be mentioned that typical yields for laboratory pyrolysis systems including our own are reported to be around 60 wt% [3,6,13]. Recovered bio-char yields were typically about 5 wt% for switchgrass and hardwood, and around 20 wt% for horse litter. This lower than usual yield of bio-char, for switchgrass and wood is due to the circulation of the bio-char back to the combustion bed via the auger rather than its elutriation form the bed and recovery via cyclone separation as would be typical for a traditional BFB pyrolysis system. Certainly, this is not the case for the horse litter but there is little explanation to offer since the ash content is similar to that of the switchgrass. One can only speculate that the solid material remaining after pyrolysis of this material is lighter and is therefore elutriated into the cyclone rather than being transported into the combustor by the auger. This needs to be further investigated in the future experiments. Having said that, the combustion of bio-char provided much of the energy to maintain the system temperatures and resulted in the observed high gas yield (~45 wt% of input biomass).

Table 4 shows how the pyrolysis vapors condensed and collected at the various points on the condensation train, along with their water

Table 2
Analysis of the biomass feedstocks used (wt%).

	Switchgrass			Hardwood			Horse Litter		
	AR	DB	DAF	AR	DB	DAF	AR	DB	DAF
C	43.6	47.79	49.10	47.07	50.56	51.07	43.2	48.14	49.30
H	5.38	5.90	6.06	5.80	6.23	6.29	4.99	5.56	5.70
N	0.46	0.50	0.51	0.17	0.18	0.18	0.35	0.39	0.40
O	39.45	43.21	44.33	39.13	42.03	42.46	39.10	43.54	44.60
Moisture	8.7			6.9			10.2		
Ash	2.38	2.60		0.93	0.998		2.12	2.37	

AR = as received, DB = dry basis, DAF = dry, ash free basis.

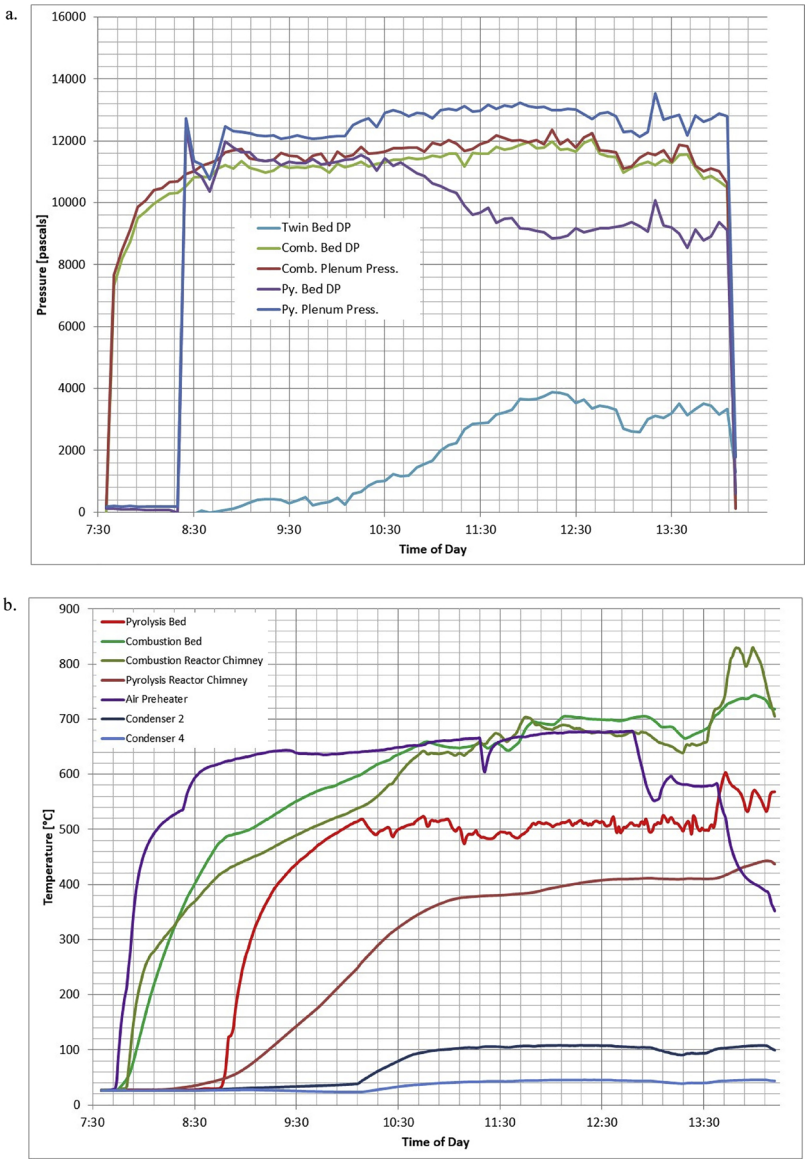


Fig. 5. Typical system pressure (a, pascals, DP = differential pressure) and temperature (b, °C) trends for heat up and biomass feeding. Shows part of the process trends including heat up (7:30 to 10:00) followed by biomass feeding beginning at 10:00.

Table 3
Product distribution for Fast Pyrolysis of Switchgrass, Hardwood and Horse Litter Feedstocks using CRIPS feeding biomass at a rate of approximately 40 kg/h.

Yields (wt%)*	Switchgrass	Hardwood	Horse Litter
Bio-oil (total liquid)	47.5 ± 6.2	49.4 ± 6.5	45.7 ± 0.3
Organic liquid	29.5	31.2	18.6
Water	17.8	18.2	27.1
Bio-char[a]	~ 5	~5	19.4
Non-condensable gas[b]	~ 45	~ 45	~35

[a] Bio-char collected at pyrolysis reactor cyclone.
[b] Non-condensable gases are calculated by difference and include pyrolysis effluent gases and an gases produced via combustion of biomass (including any bio-char or pyrolysis vapors) in the combustion chamber.
* Bio-oil yields are average of 5 runs for switchgrass; 7 runs for hardwood and 2 runs for horse litter.

Table 4
Fractionation of Pyrolysis Liquids during Fast Pyrolysis Using CRIPS.

Bio-oil fractional collection		Cond 2	Cond 3	Cond 4	ESP
Switchgrass	Fraction of bio-oil (wt%)	54.3	20.9	10.5	14.3
	Water content (wt%)	41	46	27	20
	HHV (MJ/kg)	15	15	17	20
Hardwood	Fraction of bio-oil (wt%)	56.0	20.0	11.3	12.6
	Water content (wt%)	39	46	29	20
	HHV (MJ/kg)	18	17	18	21
Horse Litter	Fraction of bio-oil	54.9	23.2	3.7	18.3
	Water content (wt%)	70	72	42	14.4
	HHV (MJ/kg)	nm	nm	mn	21.1

contents. There is little difference between the three feedstocks on how the bio-oil and water vapor fractionate within the system. The largest fraction of liquid, about half of the total, is recovered at the first collection point (labeled as Cond 2). This fraction contains about 40 wt % water. About 20% of the total liquid is collected via the next condenser (Cond 3), and this fraction contains about 46 wt% water for each feedstock. The next condenser collected about 10 wt% of the total liquid with water contents reduced to about 25–30 wt% for switchgrass and wood, but collected a less fraction for horse litter. The ESP collected

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Table 5
Analysis of ESP collected fraction of bio-oil from CRIPS and comparison with laboratory scale (process development unit, PUD) results for fast pyrolysis (N_2) and TGRP [15].

wt%	Switchgrass			Hardwood			Horse Litter
	CRIPS	PDU - N_2	PDU - TGRP	CRIPS	PDU- N_2	TGRP	CRIPS
C (db)	58.91	59.82	80.29	60.55	58.1	80.24	62.82
H (db)	5.95	6.03	5.67	6.13	6.11	5.88	5.33
O (db)	30.70	33.32	12.54	29.52	35.09	11.81	29.42
N (db)	0.38	0.92	1.5	0.45	0.7	2.07	1.52
acetic acid/ acetol	7.7	12	0.3	7.5	9.2	2.3	6.2
levoglucosan	5.0	4.5	–	4.6	9	trace	0.9
phenol/ cresols	0.7	1.1	6.3	1.0	0.6	6.7	1.4
BTEX	trace	–	4.6	0.2	0.2	3.3	0.6
naphthalene	trace	–	5.4	trace	0.2	3.4	trace

Table 6
CRIPS Condenser Bio-oil Components quantified by GC/MS.

	Cond. 2 (110 °C)			Cond. 3 (45 °C)			Cond. 4 (20 °C)		
wt%	wood	sg	hl	wood	sg	hl	wood	sg	hl
Acetic acid/ Acetol	10.98	14.02	11.0	8.99	10.14	9.8	10.95	8.97	8.4
Levoglucosan	1.39	2.03	0.2	1.59	2.74	0.2	3.24	4.07	0.3
Phenol/ Cresols	1.07	–	–	1.19	–	–	0.91	trace	0.1
BTEX	trace	–	–	trace	–	–	0.87	trace	trace
Naphthalenes	trace	–	–	trace	–	–	0.13	–	trace
Furfural	0.7	–	–	0.92	–	–	0.78	0.98	0.2
Guaiacol	0.44	–	–	0.52	–	–	0.35	0.21	0.06

sg = switchgrass, hl = horse litter.

Table 7
Composition (vol%) and higher heating value of non-condensable gases produced from Fast Pyrolysis using CRIPS and comparison with laboratory scale (process development unit, PDU) results for fast pyrolysis (N2) and TGRP [23]. Analysis is for pyrolysis effluent only, not including combustion off gases.

	CO	CO_2	H_2	CH_4	C_2H_4	C_2H_6	C_3H_8	HHV (MJ/kg)
Hardwood								
CRIPS	42.4	41.3	2.2	8.5	1.2	1.0	1.1	7.9
PDU, N_2	45.6	46.6	1.0	6.6	nm	0.0	0.0	5.47
PDU, TGRP	40.3	20.7	22.4	14.7	nm	0.9	1.0	11.36
Switchgrass								
CRIPS	41.6	34.8	2.5	7.4	1.6	0.9	0.9	8.5
PDU, N2	57.6	29.5	5.1	7.8	nm	0.0	0.0	7.7
PDU, TGRP	54.6	15.7	12.0	15.6	nm	1.1	0.8	12.4
Horse Litter								
CRIPS	34.9	39.3	3.3	12.4	1.4	1.5	0.7	7.7

Table 8
Yields from catalytic fast pyrolysis of switchgrass over HZSM-5 using CRIPS.

Product	wt% from biomass
Organic Phase Bio-oil	5.5
Aqueous Phase Liquids	8.5
Bio-char	10.0
Total Gas*	54.9
CO*	22.2
CO_2*	13.8
CH_4*	12.7
C_2H_4*	1.2

* Includes pyrolysis side gas effluent only.

about 10–18 wt% of the bio-oil, and this fraction had the lowest water content of about 20 wt%. Table 4 lists the heating values and water contents of these bio-oils collected via ESP. This fraction of the bio-oil meets ASTM D7544 (specification for pyrolysis liquid biofuel) by having a moisture content less than 30 wt%, a higher heating value (HHV) greater than 15 MJ/kg, a density between 1.1 and 1.3 kg/l, and a kinematic viscosity at 40 °C less than 125 mm^2/s [26].

Table 5 presents the elemental analysis and some major chemical constituents identified in the CRIPS ESP bio-oil fractions, and compares them to those found in similar fractions obtained from bench scale pyrolysis systems. Previous efforts on the bench scale system have realized significant thermal deoxygenation when performing TGRP i.e., pyrolysis under an atmosphere partially comprising recycled tail gas. Although the CRIPS operates with recycling of the tail gas to the pyrolysis reactor, the CRIPS bio-oil oxygen content was not as low as that obtained in the bench scale using TGRP; however, it was slightly lower than that typically achieved by traditional fast pyrolysis in the bench-scale pyrolysis system [23]. The bio-oil contains similar amounts of the major GC observable compounds to the bio-oil produced via traditional fast pyrolysis on the bench scale, but with noted decreases to the concentrations of the highly oxygenated acetic acid and acetol for both feedstocks. Trace amounts of aromatic hydrocarbons were detected in the ESP bio-oil. As shown in Table 6 the fraction collected at cond. 4 for wood had nearly 1 wt% concentration of BTEX, compounds found only using TGRP or CFP on the bench-scale system. Otherwise, the condenser fractions contained high concentrations of water soluble compounds including acetic acid and acetol along with some of the produced levoglucosan. Compositions of the major constituents identified in the producer gas (effluent from the CRIPS along with their calculated heating values are listed in Table 7 along with those from the aforementioned simple and TGRP pyrolysis reactors). Following the same trend as the bio-oil composition, it can be seen that the HHV of the CRIPS gases falls between the traditional pyrolysis and TGRP values.

5. Catalytic fast pyrolysis operation

In contrast to BFB systems, the circulating dual fluidized bed design of CRIPS allows for continuous regeneration of catalysts from coke deposits during catalytic pyrolysis. As mentioned above, three separate runs were completed using the same batch of catalysts. Each run consisted of processing about 40 kg of switchgrass over a total catalyst loading of 12.2 kg. The ratio of bed material between the pyrolysis and combustion reactors was estimated at 1/6. Feed rates were such that the weight hourly space velocity (WHSV) was about 2 h^{-1}. Achieved feed rates were lower than in the non-catalytic case, due to difficulty maintaining sufficient temperature in the pyrolysis bed at higher feederates. Table 8 presents the yields of liquid and biochar collected from the second run (presumably after catalysts have been exposed to several pyrolysis/combustion circulations) and Table 9 presents the analysis of the bio-oil collected during each run. The balance represents coke (or gas created during its combustion) and lost product as described below. In each case the bio-oil is significantly deoxygenated in comparison to that produced non-catalytically but yields of bio-oil are low. Low yields are typical for CFP, and generally the level of deoxygenation achieved and the yield of bio-oil (on a mass or carbon retention basis) are inversely proportional [27]. The oxygen contents of the bio-oil produced in these cases (particularly in runs 1 and 2) are at the lower end of those reported in the catalytic fast pyrolysis literature [17,27]. Furthermore, it is possible that the yields obtained on these trial runs with the CRIPS system may also be depressed due to gas flow problems The zeolite catalyst has a lower density than the sand used for non-catalytic pyrolysis, and therefore its use may allow a larger portion of the pyrolysis vapors to flow back into the combustion reactor, resulting in decreased liquid recovery. Collection inefficiencies for a product stream containing volatile hydrocarbons may have also contributed to the low yields. Gas yields were very high, at greater than 50% of the input

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Table 9
Analysis of Organic Fraction of ESP Bio-oil from catalytic pyrolysis of switchgrass over HZSM-5 using CRIPS.

wt%, db	Trial 1	Trial 2	Trial 3
C	80.66	86.85	74.02
H	7.67	7.92	7.38
N	0.24	0.47	0.33
O	10.64	5.3	16.4
Compound concentration (wt %)			
H_2O	1.01	0.9	2.62
BTEX	26.35	28.6	22.8
Naphthalene	2.03	1.8	1.1
Phenol	0.7	0.9	1.96
Cresols	0.9	1.36	3.5
Guaiacol	0.2	0.1	0.1
Furfural	0.02	0.04	0.70
Acetic Acid	0.05	0.02	0.8
Acetol	nd	0.2	0.2
Levoglucosan	nd	0.05	0.04

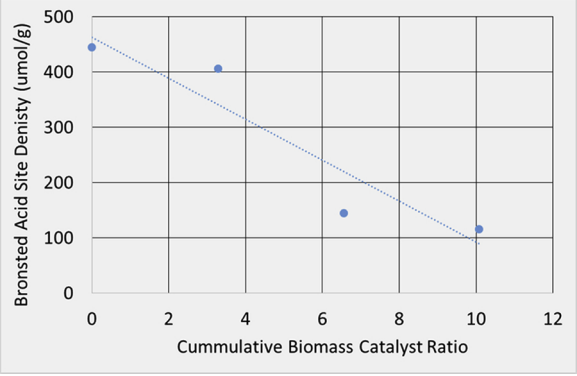


Fig. 6. Change in Brønsted Acid Site Density with use of the HZSM-5 catalyst for CFP of switchgrass in CRIPS.

biomass; CO is the major gaseous product. High gas yields are also a result of constant exposure to the most active form of the catalyst. More biochar was also collected in the catalytic pyrolysis experiments as the lower density bed material allowed for easier elutriation of the bio-char from the bed, and less was transferred to the combustion reactor via the auger. Coke on the catalyst was rather combusted making the heat typically derived from the combustion of the bio-char unnecessary. A significant amount of the catalyst (15%) was lost to elutriation of catalyst fines (over 3.25 h of operation); however, this problem could be potentially partially mitigated by design of a catalyst better suited for the flow and circulation regimes of the system. Makeup fresh catalyst was added to the system to make up for these losses prior to the next experiment.

The bio-oil collected during the first two runs was comprised largely of aromatic hydrocarbons, with little detection of oxygenated compounds. This bio-oil is much more highly deoxygenated than most reported in the literature as a consequence of the constant regeneration from coke deposits. However, the result of the third experiment was that the oxygen content of the bio-oil collected rose to 16.4 wt% and increases in the concentration of oxygenates was also detected. Analysis of the spent catalyst indicates that its Brønsted Acid site density, a key parameter for catalyst activity, was significantly decreased (Fig. 6) [28]. This is likely a result of poisoning by inorganic components of the switchgrass biomass (especially group 1 and group 2 metal ions), as has been previously reported by several studies on *in situ* catalytic fast pyrolysis [18,29,30]. However, deactivation due to dealumination resulting from exposure to high temperatures in the combustion bed cannot be ruled out.

6. Conclusion

Demonstration scale operation of a mobile trailer-mounted biomass fast pyrolysis and catalytic pyrolysis based on the combustion reduction integrated pyrolysis system (CRIPS) design was achieved. For fast pyrolysis, biomass throughput at feed rates met the standard for demonstration scale at 1 MTPD (as defined by the US-DOE) for feedstocks including switchgrass, hardwoods and waste horse litter with bio-oil yields of about 45 wt%. The organic (ESP) fractions of these bio-oils met the ASTM specifications for pyrolysis liquid biofuel. For catalytic pyrolysis over HZSM-5, continual regeneration of the catalyst from coking was achieved via circulation of the catalyst between the combustion bed and pyrolysis bed, akin to an FCC system. Bio-oils with very low oxygen content (< 10 wt%) and very high concentrations of aromatic hydrocarbons (< 25 wt%) were produced in yields of about 5 wt%. Low yields for catalytic pyrolysis maybe partially attributable to product vapor loss to the combustion bed and product collection difficulties. We continue to explore mitigation of operational challenges to improve product yield and quality and importantly increased throughput for production on-farm. While a good milestone has been reached with a successful demonstration at one metric ton per day there is still more yet to be done to get to the designed two metric ton per day operation. Longer time on stream along with measurements of effluent gas flows will be needed in the future to establish process flow simulations leading to technoeconomic analysis.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jaap.2018.11.024.

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